

2 Write a program to find the factorial of a number using recursion.

Q7 I/P

```
#include <stdio.h>
```

```
int factorial (int n) {
```

```
    if (n == 0 || n == 1)
```

```
        return 1;
```

```
    else
```

```
        return n = factorial (n-1);
```

```
}
```

```
int main () {
```

```
    int num;
```

```
    printf ("Enter a number : ");
```

```
    scanf ("%d", &num);
```

```
    printf ("Factorial = %d", factorial (num));
```

```
    return 0;
```

```
}
```

O/P

Enter a number : 10

Factorial = 3628800

main.c

Run

Share

```
1 #include <stdio.h>
2
3 unsigned long long fact(int n) {
4     if (n <= 1)
5         return 1;
6     return n * fact(n - 1);
7 }
8
9 int main() {
10     int num;
11
12     printf("Enter a number: ");
13     scanf("%d", &num);
14
15     printf("Factorial of %d is %llu\n", num, fact(num));
16
17     return 0;
18 }
```

Output

Clear

```
Enter a number: 10
Factorial of 10 is 3628800

== Code Execution Successful ==
```